

Data Flow Diagram

Date	29 June 2026
Team ID	
Project Name	Smart Lender Loan Eligibility Prediction System
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Loan Applicant → provides personal and financial details. Credit Officer / Analyst → receives prediction results for decision-making.
Rectangle with numbered header	Process	Data Collection → Applicant submits loan details via web form (Flask UI). Data Preprocessing → System validates and cleans input data. Model Prediction → ML models (Decision Tree, Random Forest, KNN, XGBoost) evaluate applicant data; best model (XGBoost) generates prediction. Result Delivery → Prediction sent to credit officer/analyst for approval/rejection.
Rectangle (solid fill, no number)	Data Store	Applicant Database → Stores applicant details (gender, marital status, education, employment,

		income, loan amount, loan term, credit history, property area). Model Repository → Stores trained ML models (XGBoost as final integrated model). Prediction Logs → Stores prediction outcomes for audit and compliance.
Labeled arrow	Data Flow	□ Applicant → (Loan details) → Data Collection. □ Data Collection → (Validated data) → Data Preprocessing. □ Data Preprocessing → (Cleaned data) → Model Prediction. □ Model Prediction → (Approval/Reject prediction) → Result Delivery. □ Result Delivery → (Decision outcome) → Credit Officer/Analyst. □ Model Prediction ↔ Prediction Logs (store results). □ Applicant Database ↔ Data Preprocessing (retrieve applicant history). □ Model Repository ↔ Model Prediction (use trained model).

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

